

Merchants of AI

Are they overvalued?

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Microeconomic Theory 584

Bubble, Bubble ... Toil and Trouble

A specter is haunting
the AI Industry – the
specter of a market
correction.



- Record investment spending (CAPEX)
- Expansions in data center construction worldwide
- Company borrowing to finance investments
- Yet market capitalization & valuation remain high.

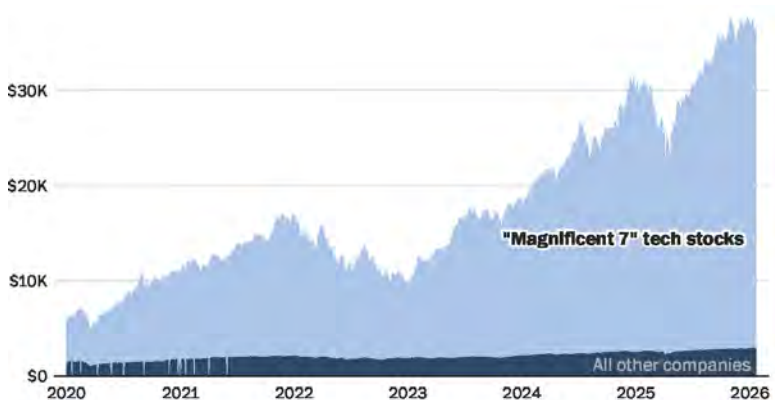
Price Earnings Ratio - Cyclically Adjusted



Source: <https://shillerdata.com>

Total Returns Index

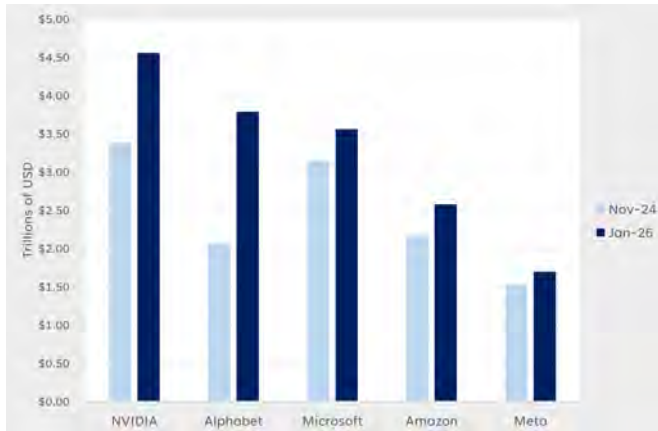
Magnificent 7: Amazon, Apple, Google, Meta, Microsoft, NVIDIA, Tesla



Source: Washington Post/Bloomberg, Feb 1, 2026

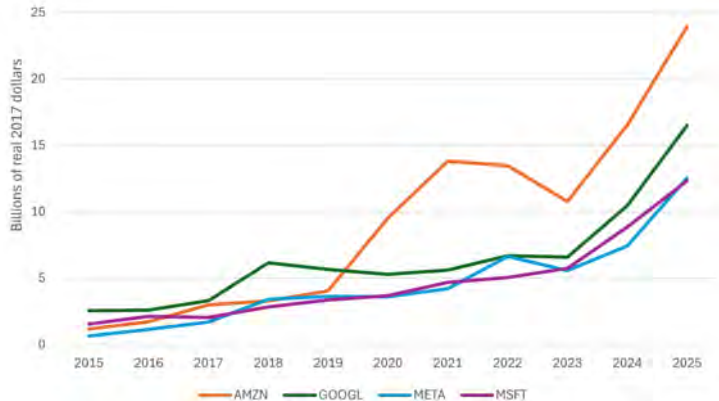
Market Capitalization

Market Valuation of Companies



Source: *Compustat*

Capital Expenditures (CAPEX)

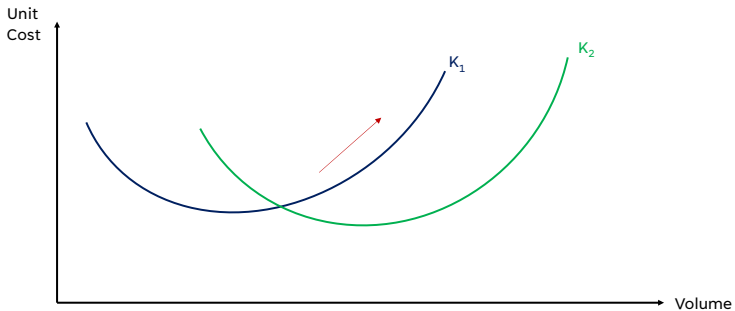


Source: Calculations based on *Compustat*

Motivation 1: Derived Demand

CAPACITY EXPANSION (DATA CENTERS)

MEET DEMAND FOR COMPUTING POWER



Motivation 2: Pursuit of AGI



- Artificial General Intelligence (AGI) is an AI capability that matches or exceeds human capabilities across a broad spectrum of cognitive tasks.
- Companies are under increased pressure to develop more **foundational** applications of AI: health care, climate change, food and agriculture, advanced robotics, & scientific discovery.

Investments in Capital and Knowledge

Company	CAPEX/Revenue		R&D/Sales	
	2024	2025	2024	2025
MSFT	0.18	0.23	0.12	0.11
META	0.24	0.38	0.27	0.28
GOOGL	0.15	0.24	0.14	0.15
AMZN	0.12	0.18	0.14	0.16
NVDA	0.02	0.02	0.10	0.09

Source: Calculations based on *Compustat*

All ratios < 1 and innovation activity appears robust

Increased Leverage

Firm	Corporate Debt (billions USD)		%Δ in Debt	Debt/Asset ratio	
	Dec 2024	Sept 2025		Dec 2024	Sept 2025
MSFT	231.2	273.3	16.7	0.43	0.43
META	93.4	109.8	16.2	0.34	0.36
GOOGL	124.2	149.6	18.6	0.28	0.28
AMZN	338.9	358.3	5.6	0.54	0.49
NVDA	32.3	42.2	26.7	0.29	0.26

Source: Annual Reports, SEC Filings

... but Debt-to-Asset ratio is fairly stable & < 1

(Note: Assets based on book value.)

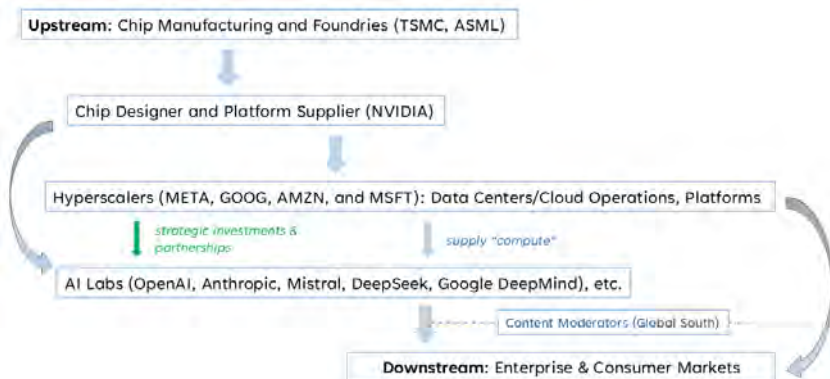
Responsible OpenAI?



- Over 900 Million+ Weekly Active Users of ChatGPT
Yet only a fraction of them are **paid subscribers**.
- Net Loss of \$5+ billion (2023-2024)
As OpenAI is a private company, no market values or operating data are available. Information is based on self-reports and news events.
- Revenue = \$3.7 billion in 2024 and \$13 billion in 2025
However, the company recorded an operating loss of \$21 billion in 2025.
- Yet CAPEX = \$9.4 bn!
OpenAI plans to spend \$1.4T more on CAPEX during 2025-2035.

Supply Chain Reaction

Global Value Chain in AI



Valuation Framework

Market Value of a Firm (V) $\propto$ Firm's Fundamentals (F)

$$V = \underbrace{\pi_0 + \frac{\pi_1^e}{(1+r)} + \frac{\pi_2^e}{(1+r)^2} + \frac{\pi_3^e}{(1+r)^3} + \dots}_{\text{Present discounted value of expected future profits } (\pi^e)}$$

where $\pi = \text{Revenue} - \text{Costs}$.

Tests are challenging, and results are mixed.

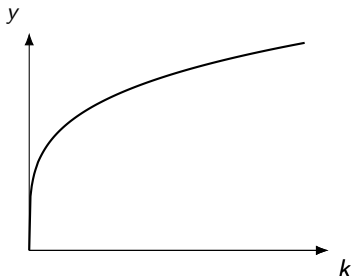
In reality,

$$V = \underbrace{\text{Fundamentals } (F)}_{\text{Economics}} + \underbrace{\text{"Noise" } (\varepsilon)}_{\text{Psychology}}$$

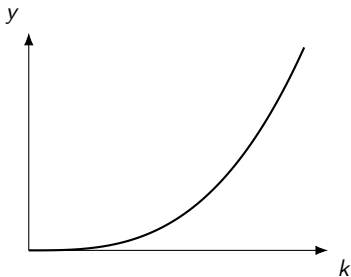
Data Center construction: Stop the Steel

Scale Economies

Diminishing returns



Increasing returns



$y = f(k)$, $y = \text{AI output}$, $k = \text{data-center compute capacity}$.

Vulnerabilities facing LLMs

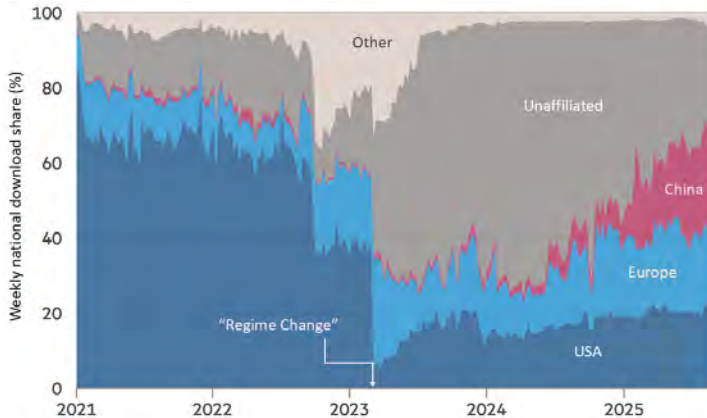
- ① Diminishing returns to scale
 - ② Resource Scarcity (water, energy) and environmental stress
 - ③ Threats to 'fair use'
 - ④ Quantum computing
 - ⑤ Competition from Open Models and China
- Let's explore this last angle further ⇒

Geopolitics

Competition with China:

- U.S. instituted export controls (Oct. 2022) on advanced semiconductors and semiconductor manufacturing equipment.
- Unintended consequences:
 - Spurred Chinese R&D. Huawei improved and boosted the production and export of its *Ascend* chips.
 - China imported older GPUs of NVIDIA. A, H series; not the B.
 - China focuses on diffusion, U.S. on innovation.
- Chinese labs 'flood' the world with their open-weight models.
Note: Open-weight $\neq$ open source.
- To facilitate adoption of their platforms, Chinese labs allow free downloads of their AI models. (Loss-leader strategy.)

China leads Model Downloads



Source: Melissa Heikkilä, *Financial Times*, Nov. 25, 2025.
Data from Hugging Face

Competition from Open Models

Examples:

Closed Models	Open Models
OpenAI ChatGPT	Llama (Meta)
Anthropic Claude	DeepSeek
Google Gemini	Mistral
xAI	Granite (IBM)

Why open innovation and why release open models?

- Personal rewards. Social community. Learning-by-doing.
- Outsource or crowdsource solutions to technical problems.
- Open models help seed adoption and diffusion, create standards.
- Developers can earn revenue from complementary assets (cloud hosting, tech support, consulting, ...)

Note: Free $\neq$ No Cost. Open models still need chips, servers, electricity.

Study by Nagel and Yue (2025)

- Dataset from *Open Router*, a platform for LLM services. Contains data on pricing and usage.
- Three key findings:
 - ❑ Open models are cheaper (one-sixth the price of closed models).
 - ❑ Open models perform almost as well as closed models, around 90%, based on ability to answer PhD-level science questions and on which responses human readers preferred.
 - ❑ Yet users predominantly use closed models, not open (about 80% of token usage is in closed models).
- Conclusion: Consumers are missing out on unrealized savings by not switching.
- Implication for financial bubbles: Open models are not yet a major threat to large proprietary companies, like Microsoft, Google, and OpenAI.

Why are Open Models Underutilized?

- Brand loyalty, switching costs (lock-ins)
- Trust and reliability. Security risks (data exfiltration and safety compliance).
- Open models are for experts, not the layperson
- Open models don't specialize in frontier innovation. They focus on making improvements and modifications, adapted to specialized needs.
- At least open models may discipline the pricing of closed, proprietary models (keep prices in check).

Prognoses: Scenarios

Is there a tech bubble? Will it burst?

- ① All Hype, Mirage.
- ② Severe Market Correction, then Recovery.
- ③ Mild/Modest Correction.
- ④ Market valuations validated (ROI realized).
- ⑤ AGI! Superintelligence. Beyond our wildest imagination.

Summary: Flowers for Algorithms?



Hinges on several factors (future 'Fundamentals')

- Economies (dis-economies) of scale
- AI adoption moves beyond experimentation and pilot stages.
- Competition from open models, foreign entities (China).
Significance for bubbles: if open models dominate, AI can flourish even as U.S. firms suffer financially.

Cleansing effect of market corrections:

- Improve resource allocation.
- Spur green innovation ("Green is all you need").
- Reduce labor market disruption.